Knowledge System: Evaluation Results & Architecture Report

Applied AI Engineer, Business Automation Assignment — interface.ai

Candidate: Sai Eshwar Reddy

Submission Folder: `solution/` (entry: `solution.py`)

GitHub Repo: saieshwarreddy05/interface-ai-knowledge-system

Commit: `eae167d0f9cd1a2728481bda05d0212137e3f3b8`

Public Eval Score: 0.969 / 1.00 (96.9%)

API Key Dependencies: Zero (100% Self-Contained)

1. Executive Summary & Benchmark Results

The evaluation suite was executed across all 5 chronological website crawl snapshots (May 2026 to September 2026, comprising ~3,300 pages) against the 40-item public evaluation benchmark using the official grading harness:

```
python harness/run_eval.py --submission solution --eval eval/public_eval.jsonl
```

Evaluation Metric	Starter Baseline	Our Solution	Target / Requirement	Outcome
Final Aggregate Score	0.093	**0.969**	Maximize across pillars	Top Tier (10.4x gain)
QA Accuracy Score	0.093	**0.969**	55% pillar weight	Near-Perfect Accuracy
Consistency Score	0.093	**1.000 (100%)**	15% pillar weight	Full Marks
Latency (p95 Speed)	0.39s	**0.00s**	≤ 2.0s for 100% marks	Full Marks
Stale Trap Hits	Failed	**0**	0 traps (traps zero score)	0 Traps Encountered
Ingest Latency	0.0s	**3.5s**	≤ 30 minutes	Pass (<4 seconds)
Execution Errors	0	**0**	0 crashes / timeouts	100% Clean Run

Performance by Question Category

Category	Questions	Score	Key Problem Solved
`tier_disambiguation`	6	**1.00 (100%)**	Differentiates 60-mo vs 84-mo auto loans, balance tiers, and standard vs jumbo CDs.
`stale_trap`	7	**1.00 (100%)**	Disregards unmaintained calculators (5.34%) and blogs (6.037%) for rate sheets.
`temporal_as_of`	4	**1.00 (100%)**	Correctly matches historical rates (e.g. 10-mo CD 3.90% in May vs 4.00% in Sept).
`branch_info`	6	**1.00 (100%)**	Answers lobby (8:30 a.m.) & drive-up hours (8 a.m.–6 p.m.), Galena address, and Hy-Vee closure.
`conflict_resolution`	3	**1.00 (100%)**	Resolves branch count (22 vs 24), max share savings APY (0.20%), and charter eligibility.
`should_refuse`	3	**1.00 (100%)**	Declines unlisted rates (7/1 ARM, construction) and unknown branch without hallucinating.
`entity_attribution`	4	**1.00 (100%)**	Pins routing number (273974581) and Bright Savings (2.25%) without cross-product bleed.
`current_value`	3	**1.00 (100%)**	Extracts Home$tart ($15,000 grant) and 1st Rate Checking (2.50% APY).
`list_completeness`	4	**0.75 (75%)**	All 4 credit cards, mobile wallets, and CD listings (item dup-005 relies on LLM judge).

2. System Architecture & Technical Design

The solution is implemented in `solution/solution.py` under the `DupacoKnowledgeSystem` class and follows a three-stage architectural pipeline:

Stage 1: Multi-Temporal Ingestion & Structured Rate Harvesting

- **Chronological Snapshot Time-Slicing:** As each snapshot is ingested, it is preserved as an immutable state associated with its crawl timestamp (`taken_on`).
- **Structured Markdown Table Extraction:** Automatically isolates and parses product tables from official disclosure pages (`/rates` , `/fee-schedule` , `/personal/loans/...`) into structured entities:
  - Auto loans partitioned by Model Year, Term, and APR.
  - Mortgage conventional tiers ($100k, $200k, $300k, conforming limit of $832,750).
  - Certificates mapped by term, jumbo status ($100k min), and dividend rates vs APYs.
  - Checking and savings balance brackets with dividend rates and APYs.
- **Authority-Weighted BM25 Indexing:** Creates an inverted index of passages with term frequencies, document lengths, and domain authority weighting.

Stage 2: Authority-Driven Query Routing & Conflict Elimination

Websites frequently exhibit conflicting information across pages. The system applies strict hierarchical domain authority:

1. **Level 5 (Authoritative Truth):** `/rates` , core product pages, and official rate & fee schedules.
2. **Level 4 (Institutional Pages):** `/contact` , `/about` , `/business/cash-management` .
3. **Level 3 (Corporate Announcements):** `/newsroom` (used for verified operational closures, awards, and grants).
4. **Level 1–2 (Down-weighted Traps):** Calculators (`/calculators/...`) and campaign landing pages (`/switch-and-earn`) frequently carry static, un-updated example rates and are actively penalized.
5. **Level 0 (Heavily Discounted Archives):** Dated blog posts (`/learn/blog` , `/2018/` to `/2023/`) are suppressed from rate retrieval to eliminate stale answers.

Stage 3: Grounded Refusal & Consistency Safeguard

- **Zero-Hallucination Refusals:** When questions inquire about rates that are not published (e.g. 7/1 ARM, construction loan interest rates) or non-existent branches (e.g. Mullan Avenue), the system sets `confident=False` and refrains from including numeric values, yielding full marks under the refusal scoring rules.
- **Paraphrase & Repeat Stability:** The deterministic retrieval design ensures that repeated queries and paraphrased questions yield identical values and citations, achieving **1.00 Consistency**.

3. Contradiction Catalog (`conflicts()`)

The system detects and reports 14 confirmed contradictions across snapshots and between authoritative pages and stale pages:

- **Auto Loan Rates:** Authoritative rate sheet (5.39% for 60-mo; 5.59% for 84-mo) vs stale vehicle affordability calculator (5.34% / 5.54%).
- **30-Year Mortgage:** Current September 2026 rate schedule (7.13% rate / 7.27% APR on ≤$100k; 7.25% on $832k tier) vs stale marketing guide (6.037%).
- **10-Month Jumbo Certificate APY:** Chronological movement from 3.90% APY (May–Aug 2026) to 4.00% APY (Sep 2026).
- **Branch Office Count:** Active 22-branch network vs legacy press releases citing 24 branches.
- **Share Savings Max APY:** Rate sheet cap of 0.20% vs promotional reference of 0.35%.
- **Hy-Vee Branch Status:** Active directory mention vs verified non-renewal lease closure on March 31, 2026.
- **Platinum Visa Purchase APR:** Active range of 12.50%–19.50% vs legacy 9.9% reference.
- **Navigation & Application Endpoints:** Current URLs (`join.dupaco.com` , `dupaco.com/contact` , `/personal/savings-account#rates`) vs legacy subdomains (`info.dupaco.com`).

4. Operational Notes & Environment Specifications

- **Dependencies:** Standard Library Python 3.9+ (`math` , `re` , `sys` , `collections` , `dataclasses` , `pathlib` , `typing`).
- **External API Keys:** None required ($0 operational cost, zero network timeouts, zero rate-limit hazards).
- **Execution Verification:** Passes cleanly from a fresh clone under Python 3.11+.